

PHLOP: Physics-Grounded Hierarchical Latent space for Object state and Presentation*

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0009-0004-2795-6739

Abstract—Understanding how objects move, interact, and respond to physical forces remains a key challenge in AI. Existing benchmarks such as CLEVRER, COMPHY, and Physion++ either rely on symbolic labels or lack interpretable grounding for physical dynamics. We introduce PHLOP (*Physics-grounded Hierarchical Latent space for Object state and Presentation*), a diagnostic benchmark that bridges low-level physics and high-level reasoning. PHLOP uses MuJoCo simulations to procedurally generate diverse scenes with realistic object parameters (e.g., mass, friction, elasticity), and augments them with a hierarchical annotations defined by a formal taxonomy, covering different levels of abstraction of physical events (e.g., rolling, sliding, collision). Each video is annotated per-object, per-frame resolution. A question-generation module produces difficulty-graded queries spanning temporal, causal, and material reasoning over domain-generalization (out-of-distribution) splits. Experiments with open video-language models reveal substantial gaps in zero-shot physical understanding; parameter-efficient (LoRA) fine-tuning across difficulty curricula closes much of this gap, while textual physics hints help unevenly across models. PHLOP enables interpretable, reproducible, and fine-grained evaluation of physical reasoning and supports future research at the intersection of simulation and language.

Index Terms—simulation, physics engine, taxonomy, object properties, annotation, video question answering, visualization

I. INTRODUCTION

Physical reasoning, which involves understanding and predicting how objects interact, move, and respond to forces, is fundamental to both human cognition and machine intelligence (Yi, Wu, Gan, Torralba, Kohli & Tenenbaum 2020, Tung, Ding, Chen, Bear, Gan, Tenenbaum, Yamins, Fan & Smith 2023, Battaglia, Hamrick & Tenenbaum 2013, Piloto, Weinstein, Battaglia & Botvinick 2022a). From an early age, humans intuitively grasp complex physical interactions (Piloto et al. 2022a, Battaglia et al. 2013). For instance, when watching a ball bounce off the floor, people naturally expect it to slow down or bounce back depending on its elasticity, even without explicit calculations (Battaglia et al. 2013, Piloto et al. 2022a). Despite rapid advances in deep learning, AI models continue to struggle with structured and interpretable physical reasoning (Tung et al. 2023, Battaglia et al. 2013, Piloto et al. 2022a, Melnik, Schiewer, Lange, Muresanu, Saeidi, Garg & Ritter 2023, Wang, Li & et al. 2024). Bridging this gap is crucial for applications in robotics, computer vision,

and AI, where models must interpret dynamic scenes and make physically plausible predictions (Yi et al. 2020, Tung et al. 2023, Battaglia et al. 2013).

Existing benchmarks, such as CLEVRER (Yi et al. 2020), COMPHY (Chen, Yi, Li, Ding, Torralba, Tenenbaum & Gan 2022), and Physion++ (Tung et al. 2023), have significantly advanced our ability to evaluate models on tasks of physical reasoning; however, each comes with its limitations. CLEVRER relies on symbolic labels for events (e.g., “collision”) that lack explicit physical grounding, making it challenging to analyze fine-grained dynamics (Yi et al. 2020). COMPHY enhances interpretability through compositional physical concepts while simplifying physics by excluding rotational dynamics and approximating friction modeling (Chen et al. 2022). Physion++ integrates a wide range of physical properties and scene diversity, yet it lacks per-frame annotations and event-level semantic labeling, limiting interpretability for temporal and causal analysis (Tung et al. 2023).

Recent work highlights this disconnect, as models struggle to infer latent physical properties from purely visual cues (Hofherr, Sorrenson, Li, Torralba, Schölkopf & Stückler 2023), or match human-level physical reasoning (Piloto & et al. 2024). For instance, while a physics engine may simulate a bouncing ball using precise coefficients, humans describe the same event in qualitative terms: “the ball bounces high because it is very elastic” (Battaglia et al. 2013, Piloto et al. 2022a).

To address these limitations, we introduce **PHLOP** (Physics-Grounded Hierarchical Latent space for Object state and Presentation), a novel benchmark and data set developed from the ground up. PHLOP unifies detailed physical simulation with structured semantic annotations to support hierarchical reasoning over dynamic physical events. Our key contributions include:

- **Physics-grounded Dataset:** PHLOP is a novel benchmark that provides per-object, per-frame data generated from rigid-body simulations. It includes rich physical attributes (e.g., mass, velocity, friction, elasticity) and frame-aligned annotations of physical events grounded in continuous dynamics.
- **Hierarchical physical taxonomy:** We design a structured taxonomy that captures multiple levels of abstraction,

TABLE I

COMPREHENSIVE COMPARISON OF PHYSICAL REASONING BENCHMARKS. PHLOP UNIQUELY OFFERS DETAILED PER-FRAME ANNOTATIONS DEFINED BY FORMAL TAXONOMY, EXPLICIT FRICTION AND ROTATIONAL DYNAMICS MODELING, AND INTERPRETABLE PHYSICAL EVENT DETECTIONS. BOLD ROWS HIGHLIGHT CAPABILITIES THAT ARE UNIQUELY OR FULLY SUPPORTED BY PHLOP. ✓= FULLY SUPPORTED, -= PARTIALLY SUPPORTED, ✗= NOT SUPPORTED.

Capability / Dataset	PHLOP	Physion++	COMPHY	CLEVRER	ContPhy	AGQA	CATER	PHYRE
<i>General Capabilities</i>								
Question Answering	✓	✓	✓	✓	✓	✓	✓	✗
Diagnostic Annotation	✓	-	-	-	-	✗	-	-
Per-frame Annotations Defined by a Formal Taxonomy	✓	✗	-	✗	✗	✗	✗	✗
Counterfactual Dynamics	✓	-	-	✓	-	✗	✗	✗
Explicit Physical Properties	✓	-	-	✗	-	✗	✗	-
Few-shot Material Reasoning	✓	-	-	✗	✗	✗	✗	✗
<i>Fine-Grained Physical Reasoning</i>								
Real-valued Properties (mass, friction, etc.)	✓	-	-	✗	-	✗	✗	✗
Rotational Dynamics	✓	✗	✗	✗	✗	✗	✗	✗
Friction-induced Events	✓	✗	✗	✗	✗	✗	✗	✗
Elastic vs. Inelastic Collision	✓	-	-	✗	✗	✗	✗	✗
Moving Event Onset Detection	✓	✗	✗	✗	✗	✗	✗	✗
Distribution-based Material Sampling	✓	-	-	✗	✗	✗	✗	✗

from low-level motion and contact dynamics to high-level events, enabling interpretability and flexible supervision.

- **Semantic-Physical Alignment:** PHLOP maps numerical physical behaviors to interpretable labels, bridging the gap between low-level physics and high-level reasoning tasks.
- **Domain-generalization splits:** PHLOP is partitioned into TRAIN/VAL/TEST sets whose generative factors (shapes, materials and their physical sub-components, object counts, collision modes, floor textures, and camera profiles) are deliberately disjoint, so evaluation measures out-of-distribution generalization rather than memorization.
- **Difficulty-graded questions:** Each generated question is annotated with a difficulty label (*easy/medium/hard/very_hard*) reflecting the depth of reasoning required, enabling curriculum-based analysis.
- **Model evaluation and learning analysis:** We evaluate open video-language models both zero-shot and under LoRA fine-tuning, with and without textual physics hints, characterizing how curriculum composition and non-visual cues shape physical reasoning.
- **Reproducible Simulation Framework:** The project includes a GitHub repository that enables the generation of new scenes with controllable physics. PHLOP provides a reproducible method for creating and recreating specific scenes with various object properties and conditions.

II. RELATED WORK

Recent years have seen significant progress in developing datasets that support the study and evaluation of machine un-

derstanding of object dynamics, causal structure, and intuitive physics in videos (Yi et al. 2020, Tung et al. 2023). These benchmarks can be categorized as (i) symbolic diagnostic benchmarks, (ii) physics-grounded simulations, and (iii) video-based reasoning task. We briefly review each and position PHLOP within this landscape.

A. Symbolic Diagnostic Benchmarks

The early dataset, such as CLEVR (Johnson, Hariharan, van der Maaten & et al. 2017) and CLEVRER (Yi et al. 2020), primarily focused on symbolic or template-based reasoning. CLEVR (Johnson et al. 2017) established a structured visual question-answering (VQA) system involving object attributes (e.g., “Is there a red cube to the left of the blue sphere?”) using photos of synthetic scenes, but due to its nature, it lacked temporal or physical dynamics. CLEVRER (Yi et al. 2020) was built on this idea and pioneered video-based causal and counterfactual question answering, around object collisions and motion on the surface. For instance, “What caused the red ball to stop?” or “What would happen if the green object were removed?”. However, physical interactions are abstracted into high-level symbolic labels (e.g., “collide”) without grounding in continuous physical parameters, limiting their utility for fine-grained physical analysis (Yi et al. 2020). Answers to diagnostic questions remain symbolic, lacking annotations that are defined by a formal taxonomy and cover different levels of abstraction, or explicit physical properties to explain causality. IntPhys (Riochet, Castro, Bernard & et al. 2018) introduced violations of intuitive physics, and CRAFT (Wu, Jiang, Sun & et al. 2020) incorporated causal reasoning with forces, but

both rely on categorical abstractions rather than physically grounded measures.

B. Physics-Grounded Simulations

To move beyond symbolic abstraction, recent benchmarks have incorporated physics engines to generate realistic dynamics. PHYRE (Bakhtin, van der Maaten, Johnson & et al. 2019) provides a library of 2D physical puzzle scenes governed by gravity, where tasks involve predicting outcomes under known initial conditions. However, PHYRE does not expose object-level physical parameters such as friction or mass.

ComPhy (Chen et al. 2022) expands the scope to more complex 3D multi-object dynamics, supporting compositional physical reasoning (e.g., removing an object and re-simulating). However, it has limited material diversity, representing materials with categorical labels, and simplifies physics by assuming uniform friction and omitting rotational dynamics and distribution-based sampling of physical parameters.

Physion++ (Tung et al. 2023) further extends these approaches, introducing complex interactions and scene diversity by varying object properties (e.g., size, shape, density) and simulation types (rigid, fluid, deformable). It focuses on high-level prediction tasks (e.g., “Will the object fall off the table?”) rather than detailed physical analysis, and lacks structured semantic grounding of events or per-frame physical state labels (Tung et al. 2023).

Recent approaches like SuperCLEVR (Mao, Gan & Tenenbaum 2024) and PhysGen (Fang & et al. 2024) explore procedural video generation and physics-based prediction, but do not provide per-frame, object-level semantic labels as PHLOP does. InfLevel (Piloto, Weinstein, Battaglia & Botvinick 2022b) measures model surprise under physically implausible conditions (e.g., gravity violation), offering developmental psychology-inspired diagnostics but not per-object tracking.

ContPhy (Zhang, Chen, Zhao & Fan 2024) proposes attribute-based scene analysis (e.g., object softness or size) in physically consistent environments. While this supports high-level reasoning, it does not offer per-frame semantic annotations. Hofherr et al. (Hofherr et al. 2023) explore real-valued parameter estimation using neural implicit functions, and Liu et al. (Liu & et al. 2024a) frame physical parameter estimation as an inverse-graphics problem. Both works highlight the need for interpretable, structured metadata like that in PHLOP.

C. Video Reasoning Datasets

Some datasets approach physical reasoning as part of general video understanding. CATER (Girdhar & Ramanan 2019) focuses on object permanence and spatial tracking under occlusion, but uses simple 3D object shuffles without realistic physics. AGQA (Grunde-McLaughlin, Krishna & Agrawala 2021) scales video QA to real-world scenes, asking about action sequences and object interactions, but relies on high-level semantic cues rather than low-level physical simulation. While these datasets advance multi-modal understanding, they are

not designed to explore questions requiring physical causality, material properties, or dynamical system behavior.

More recent efforts such as Helping Hands (Cheng, Das & Hauptmann 2023), Learning Object States (Sun & et al. 2023), and disentangled SDE dynamics (Lee, Li, Gan & Tenenbaum 2023) contribute to object-centric video reasoning, highlighting temporal dependencies and state changes in visual data. While these datasets advance multimodal understanding, they are not designed to explore questions requiring physical causality, material properties, or dynamical system behavior—domains where PHLOP aims to bridge the gap.

D. PHLOP in Context

Recent work leverages procedural priors for scene generation and video prediction (Takenaka, Ma, Chang, Santoro & Mirowski 2023), suggesting that control over simulation parameters can guide model learning. The design of PHLOP is also inspired by theoretical foundations: simulation-based cognition (Battaglia et al. 2013), intuitive physics learning (Piloto et al. 2022a), and physics-guided machine learning (Karniadakis, Wang & Pang 2023). Reference material properties are grounded in established engineering and physics resources (Ashby & Jones 2018, Callister 2020, Halliday, Resnick & Walker 2013), ensuring physical realism in the dataset. PHLOP builds on these advances while addressing critical limitations in existing physical reasoning datasets. Table I outlines how the dataset uniquely bridges low-level dynamics and high-level semantics.

PHLOP uniquely enables:

- **Per-frame semantic labeling** of object states and transitions—absent in all prior benchmarks.
- **Rotational and frictional event detection**, such as rolling, slipping, or friction-induced halts.
- **Elastic vs. inelastic collision analysis** using velocity changes and energy loss metrics.
- **Few-shot material reasoning** via continuous distribution-based sampling across density, elasticity, and friction.
- **Moving event onset detection**, allowing frame-accurate identification of motion transitions.

These capabilities enable novel types of questions such as:

- “Which object experienced the greatest kinetic energy loss?”
- “When did the red ball stop rolling?”

Such queries require the unified physical and semantic grounding that prior datasets lack. By combining continuous dynamics with interpretable labels, PHLOP offers a scalable, diagnostic benchmark for evaluating grounded physical reasoning in vision-language models.

III. METHODOLOGY

In this section, we describe the creation of PHLOP, which involves two core components: (1) the dataset creation process for generating physics-accurate videos with ground truth annotations, and (2) the benchmark evaluation framework for assessing video-language models on physical reasoning tasks.

A. Dataset Creation

1) *Video Generation Process*: PHLOP contains roughly 20,000 generated videos - partitioned into 16,000 TRAIN, 2,000 VAL, and 2,000 TEST scenes (Section III-A4)- each depicting a simulated physical scene involving multiple rigid objects with varying properties and interactions.

Scenes are created using the MuJoCo physics engine (Todorov, Erez & Tassa 2012). Each scene involves one or more physical events such as collisions, sliding, rolling, or frictional deceleration. The simulation captures these interactions at a high resolution of 1024×768 pixels at 25 FPS, capturing dense temporal sequences of interactions. This object-centric logging bridges physical reasoning and visual understanding by aligning data with physical ground truth.

The dataset is generated using a controlled randomization method over object attributes (shape, size, color), physical properties (e.g., friction and restitution), and initial conditions (e.g., velocity vectors, spatial configurations).

2) *Semantic Annotation and Taxonomy*: Physical events within each video are annotated automatically at the frame and object level using PHLOP’s hierarchical taxonomy. This taxonomy categorizes interactions across several abstraction levels:

- **Level 1**: Broad physical category (e.g., Kinematic Events)
- **Level 2**: Event type (e.g., Rotational Motion)
- **Level 3**: Fine-grained label (e.g., Rolling with Slipping)

Table II defines this taxonomy. It allows PHLOP to support taxonomy-based prompting and structured question generation.

Annotations are derived from simulation logs using rules over motion thresholds, velocity changes, and force-based triggers. For example, an object labeled as “Friction Stop” must meet criteria for velocity approaching zero under non-collisional deceleration. Collisions are further graded by the fraction of kinetic energy retained, yielding *Elastic*, *Partially Inelastic*, and *Highly Inelastic* labels rather than a single binary category.

Shape-aware detection. Event detection is conditioned on object geometry, since the same kinematic signature implies different physical events for different shapes. Spheres are eligible for rotational labels (rolling, slipping) but are excluded from sliding-friction detection, whereas cubes and blocks are treated as sliding contacts. Cylinders are handled according to their instantaneous orientation: an *upright* cylinder (resting on a flat end) is evaluated for sliding-friction events, while a cylinder lying on its side is evaluated for rotational motion (rolling and rolling-with-slipping).

3) *Parameter Sampling*: To ensure both visual and physical diversity, PHLOP samples a range of object-level and global scene parameters. Tables III and IV summarize the full range of material and scene properties used.

TABLE III
MATERIAL PHYSICAL PROPERTIES WITH REAL-WORLD REFERENCE VALUES (ASHBY & JONES 2018, CALLISTER 2020).

Material	Elasticity (–)	Density (kg/m ³)	Friction (–)	Examples
Metal	0.85–0.92	2700–19320	0.28–0.32	Steel, Aluminum, Copper
Wood	0.35–0.45	500–700	0.45–0.55	Softwood, Hardwood
Rubber	0.90–0.98	900–1200	0.90–1.10	Natural, Synthetic
Glass	0.55–0.65	2400–2600	0.18–0.22	Soda-lime, Borosilicate
Plastic	0.65–0.75	1000–1800	0.35–0.45	LDPE, PP, PC

Material Physical Properties: Table III presents material-specific property ranges with representative real-world context. Each object in the simulation is assigned properties that determine its physical behavior and visual appearance. Sampling was performed from the material-specific distributions that can be found in Table III. Each object is assigned:

- **Shape**: Chosen randomly from sphere, cube, cylinder, or block, each with different moments of inertia and contact surfaces.
- **Material Type**: Samples randomly from a discrete set (e.g., wood, metal, rubber, glass, plastic), each associated with different parameter distributions.
- **Physical Properties**: Material-dependent distributions for density, elasticity, and friction (Table III).
- **Visual Properties**: Material-specific transparency, specular, shininess, and random color.
- **Dimensions**: Ball/cylinder radius 0.05–0.1m, cube side 0.05–0.1m, block 0.01–0.1m $\times$ 0.05–0.1m $\times$ 0.05–0.1m.
- **Initial Position & Orientation**: Randomized in 3D space to create spatial variability across simulations.
- **Mass**: Calculated from volume and sampled density, minimum 1e-6 kg.
- **Initial Velocities**: Linear velocities are sampled in meters per second (m/s) and angular velocities in radians per second (rad/s). They uniformly sampled in 3D space $[-1, 1] \times [-1, 1] \times [-1, 1], (x, y, z)$.

Global Scene Parameters: Table IV provides comprehensive specifications for environmental factors:

- **Lighting**: *Position*, *color*, *reflectivity*, and *angular spread* control the directionality and softness of shadows and highlights.
- **Camera**: *Azimuth* and *elevation* define the viewing angle; *distance* and *look-at point* control framing and focus.
- **Floor Plane**: *Friction coefficient*, *color* (based on friction), *shininess*, and *specular reflectance* are sampled to vary how objects interact with surfaces visually and physically.

All parameters—both global and per-object—are stored alongside the simulation for full reproducibility and analysis. These attributes drive the emergence of behaviors such as gliding, bouncing, tumbling, and frictional halts, allowing PHLOP to capture a comprehensive range of Newtonian phenomena.

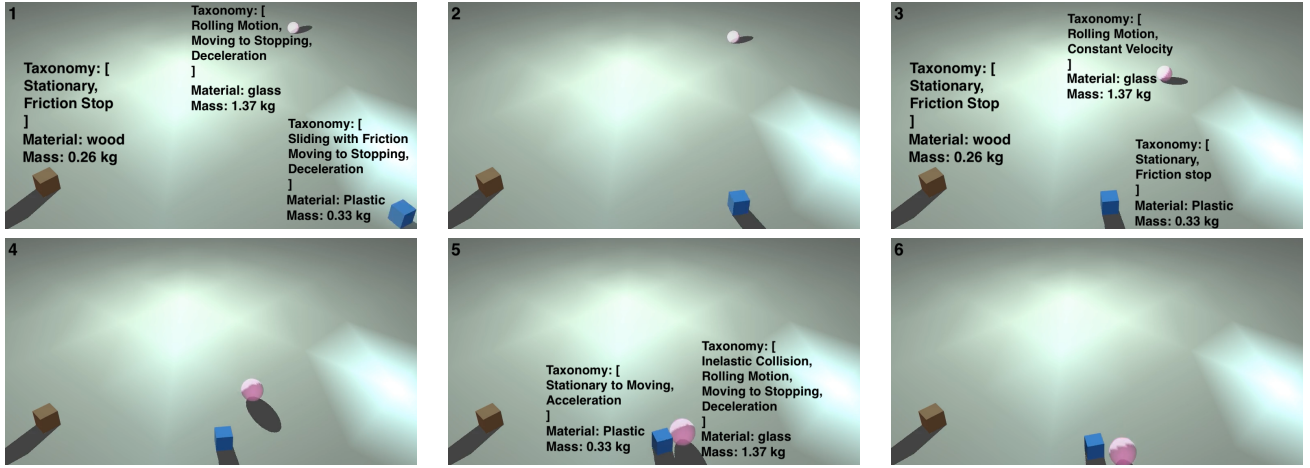


Fig. 1. Sequence of 6 frames showing dynamic object interactions and taxonomy-based annotations in the PHLOP simulation. Each frame captures per-object motion, material properties, and physical events.

TABLE II

HIERARCHICAL TAXONOMY OF PHYSICAL EVENTS IN PHLOP. EVENTS ARE ORGANIZED INTO GENERAL CATEGORIES (LEVEL 1), SUBCATEGORY TYPES (LEVEL 2), AND SPECIFIC EVENT LABELS (LEVEL 3). COLLISIONS ARE GRADED INTO THREE ENERGY-BASED CLASSES; ROTATIONAL AND FRICTION EVENTS ARE DETECTED WITH SHAPE- AND ORIENTATION-AWARE RULES.

Level 1 (General Category)	Level 2 (Event Type)	Level 3 (Specific Event Labels)
Kinematic Events	Linear Motion	Constant Velocity Decelerating Accelerating Stationary
	Rotational Motion	Pure Rotation Rolling Motion Rolling Motion with Slipping
Interaction Events	Collisions	Elastic Collision Partially Inelastic Collision Highly Inelastic Collision
State Transitions	Motion Change	Moving to Stopping Stationary to Moving
Environmental Interactions	Friction-Induced Events	Friction Stop (Stopped Due to Friction) Sliding with Friction

4) *Dataset Splits for Domain Generalization:* To evaluate *generalization* rather than memorization, PHLOP is partitioned into TRAIN, VAL, and TEST splits whose generative factors are deliberately *disjoint*. Rather than randomly partitioning a single pool of scenes - which would let a model exploit shared textures, materials, or viewpoints - each split draws from a different region of the scene parameter space: distinct object shapes, materials, per-material physical sub-components (e.g. different metal alloys), object-count ranges, collision modes, floor textures, and camera profiles. Consequently, a model is tested on material–shape–viewpoint combinations it never saw during training (e.g. rubber and block objects under an “extreme” camera profile appear *only* in the test split). Table V summarizes the configuration. Because the sub-components are also disjoint, even shared material categories (e.g. glass) use different physical-parameter regions across splits. Every scene is additionally rendered under two **camera modes**—

a static fixed-viewpoint render and a moving (orbiting) render—allowing the same physical event to be probed under different visual coverage. All experiments reported here use the `static` mode for a controlled, single-viewpoint comparison; one trade-off of this fixed viewpoint is that some events can fall outside the frame or behind an occluder, rendering a fraction of questions unanswerable from the static view (quantified and handled in Section IV).

The figures 1 and 2 illustrate different scenes from the dataset, where the number of objects and environmental conditions vary. Notably, the camera position and lighting conditions differ between the two scenes. The taxonomy annotations also reflect different physical interactions. Figure 1 shows a more complex scenario involving an inelastic collision, where a moving object interacts with a stationary one, causing it to move. In contrast, Figure 2 presents a simpler case, where two objects slide and eventually stop due to friction.

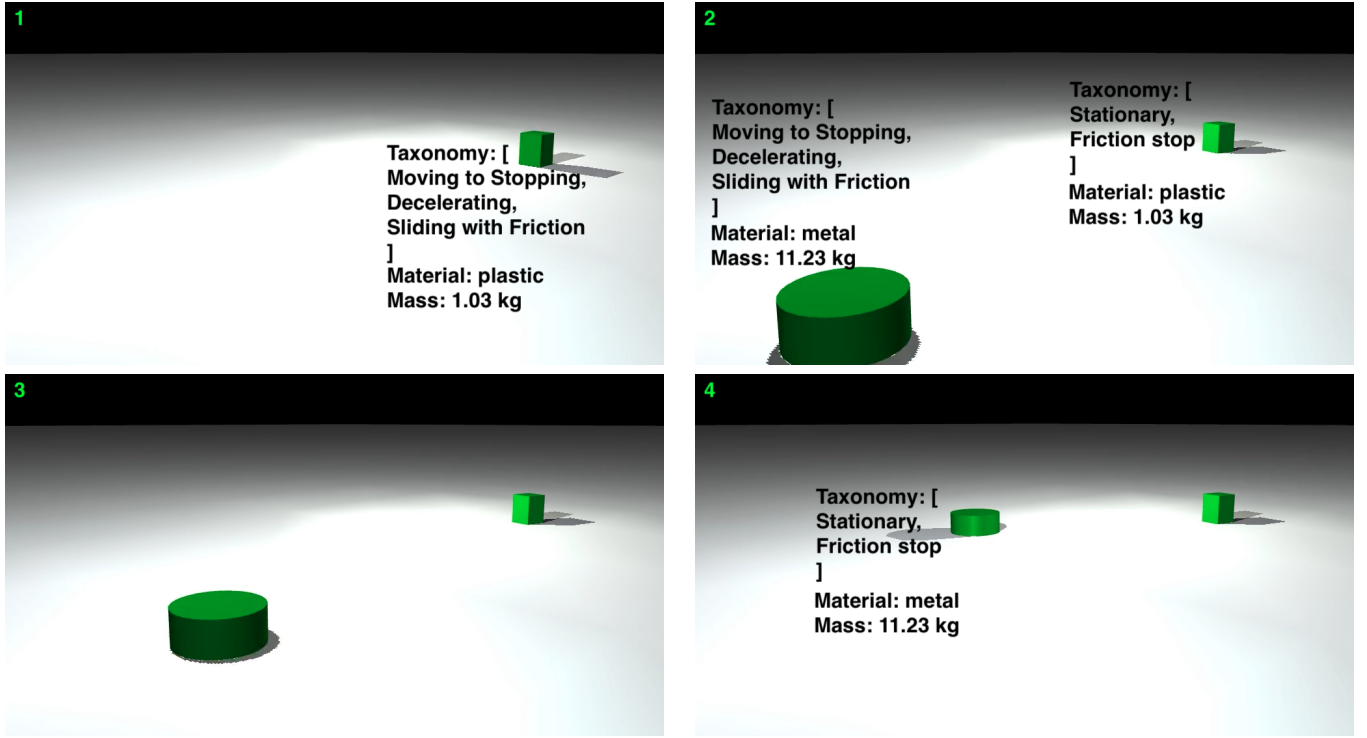


Fig. 2. Sequence of 4 frames showing dynamic object interactions and taxonomy-based annotations in the PHLOP simulation.

TABLE IV
GLOBAL SCENE PARAMETER SPECIFICATIONS

Component	Parameter Range	Description
Camera	Lookat: $[-0.5, 0.5] \times [-0.5, 0.5] \times [-0.5, 0.5]$, Distance: 1.5–3.5 m	Camera position in 3D space
Lighting	Azimuth: $[-90^\circ, -30^\circ]$, Elevation: $[-30^\circ, 0^\circ]$	Viewing angle constraints
	Position: $[-1, 1] \times [-1, 1] \times [0.1, 0.7]$	Lighting position in 3D space
	Diffuse: 0.7–1.0, Specular: 0.5–1.0	Light intensities
Floor	Cutoff: 50° – 180°	Angular spread
	Friction: Static $[0.0, 1.0]$, Dynamic $\leq$ Static	Three-component friction
	Specular: 0.2–0.8, Shininess: 0.2–0.8	Visual settings
	RGBA: brightness	Color mapping

5) *Verification and Quality Assurance*: To ensure visual-semantic alignment and correctness of the annotations, each simulation produces three video modalities: RGB video, segmentation video (with object-specific masks), and annotated video (with bounding boxes, physical attributes, and semantic event labels). These were used to manually verify that physical events, object identities, and interaction dynamics were captured correctly across time. While most annotations are generated algorithmically from simulation logs using rule-based logic, a sampled subset of the generated dataset was manually

TABLE V
DISJOINT GENERATIVE FACTORS PER SPLIT, ENFORCING OUT-OF-DISTRIBUTION EVALUATION. MATERIAL SUB-COMPONENTS (E.G. METAL ALLOYS) ARE ALSO DISJOINT ACROSS SPLITS, SO EVEN SHARED MATERIAL CATEGORIES USE DIFFERENT PHYSICAL-PARAMETER REGIONS.

Factor	Train	Val	Test
Shapes	ball, cube, cyl.	ball, cube	block, cyl., cube
Materials	metal, wood, plastic	metal, wood, glass	glass, rubber
#Objects	2–6	2–5	3–7
Camera	broad	narrow	extreme
Collisions	coll./slide/stat.	coll./slide	coll./offset/slide
Floor	checker/flat	flat/gradient	checker/gradient

reviewed to validate consistency. For example, rolling events were cross-checked against angular and linear velocity curves, or stationary objects were checked whether linear velocity is approximately zero within a threshold. Future work may explore using learned classifiers to automate this verification at scale.

B. Benchmark Methodology

1) *Automatic Question Generation*: PHLOP includes a built-in module for automatic question generation based on the annotated physical interactions and object properties. This module leverages simulation logs (mass, velocity, position, etc.) and per-frame taxonomy labels to create grounded question-answer pairs. Questions are designed to test a spectrum of reasoning skills, including:

- **Temporal Reasoning:** e.g., “At what time does the red ball become stationary?”
- **Material Property Inference:** e.g., “Which object had the highest friction coefficient?”
- **Interaction and Outcome:** e.g., “Which object was involved in the most collisions?”
- **Aggregate Events:** e.g., “How many objects rolled during the video?”
- **Energy & Loss:** e.g., “What percentage of system kinetic energy was lost in a collision?”

Unlike symbolic QA in CLEVRER (Yi et al. 2020), PHLOP’s questions are grounded in both semantic labels and physical data. This allows richer evaluation of model understanding across symbolic, causal, and physical domains. The questions and the respective correct answers are unambiguous because of their grounding in the simulation data, enabling evaluation of both the language and the compositional physical understanding.

2) *Question Difficulty Annotation:* Each automatically generated question is tagged with a **difficulty** label—*easy*, *medium*, *hard*, or *very_hard*—reflecting the depth of reasoning required rather than surface form. The label is assigned by the question generator from the underlying reasoning type:

- **Easy:** single-step perception and counting (object count, rolling detection, stopped-object count, collision presence).
- **Medium:** property comparison and attribute inference (e.g. highest-friction object, velocity/friction scaling).
- **Hard / very_hard:** multi-step physical reasoning over continuous dynamics (momentum/energy conservation, multi-hop causation, temporal-event ordering).

These labels are the basis of the fine-tuning curriculum in Section III-B4.

3) *Prompt Construction and Types:* Across both zero-shot and fine-tuning evaluation, each model receives 32 uniformly sampled frames, an instruction on how to answer, and a question with multiple-choice options, under one of two prompting settings:

- 1) **No physics hint:** video and question only; no metadata is included.
- 2) **With physics hint:** object-specific physical parameters are appended to the prompt, such as:

“1. *darkcyan sphere: mass = 0.8, friction = 0.92*”

Critically, the physics hint provides contextual cues but not the correct answer; it lists parameters for *all* objects—including those irrelevant to the current question—introducing distractors so that the model must focus on the relevant information and reason compositionally rather than retrieve a fact. Comparing the two settings isolates how much each model relies on textual physical cues versus grounded visual dynamics, in the spirit of the structured-context analyses of Physion++ (Tung et al. 2023) and ContPhy (Zhang et al. 2024).

4) *Fine-tuning Protocol and Difficulty Ablation:* We **LoRA-fine-tune** three open vision–language models—**SmolVLM**, **Qwen2-VL**, and **InternVL3** - on PHLOP, to see

TABLE VI
THE FOUR FINE-TUNING SETUPS. EACH ONE TRAINS ON A DIFFERENT SET OF QUESTION DIFFICULTIES. THE “VALIDATION DIFFICULTIES” (THE ONES LEFT OUT OF TRAINING) ARE USED ONLY TO DECIDE WHEN TO STOP TRAINING. EVERY SETUP IS THEN TESTED ON THE SAME TEST SET, WHICH INCLUDES ALL DIFFICULTIES.

Curriculum	Train difficulties	Validation difficulties
<i>easy</i>	easy	medium, hard, very_hard
<i>easy_medium</i>	easy, medium	hard, very_hard
<i>hard</i>	hard	easy, medium, very_hard
<i>full</i>	all	— (no holdout)

how the mix of question difficulties used for training affects physical reasoning. We train four versions of each model that differ only in which difficulties they are trained on (Table VI): *easy*, *easy+medium*, *hard*, or *full* (all difficulties). While a model is training, we track its accuracy on the difficulties it was *not* trained on and use that to decide when to stop. We then test every model- and the original zero-shot model -on the test set, and this test set includes questions of *all* difficulties. In other words, the curriculum name only says what a model was *trained* on; all models are *tested* on the same complete test set (Section IV). This lets us measure two kinds of generalization: to new scenes (new materials, shapes, and camera views; Table V) and to question difficulties the model never saw in training (Table VI).

Each curriculum is run under the two prompting settings of Section III-B3 (no physics hint / with physics hint). Together with the non-fine-tuned *zero-shot* baseline, this yields five conditions per model and prompting setting, all evaluated on the same held-out TEST split of 25,436 questions (a fraction of which are unanswerable from the static viewpoint; Section IV). Results are reported in Section IV.

IV. RESULTS

We report accuracy exactly as produced by the official PHLOP evaluator on the held-out TEST split. For every model we compare five conditions—the non-fine-tuned *zero-shot* baseline plus the four LoRA curricula of Table VI (*easy*, *easy+medium*, *hard*, and *full/all-tasks*)—each under the two prompting settings of Section III-B3 (no physics hint / physics-augmented). Beyond a single overall score, the evaluator also reports accuracy separately for each *test-question difficulty* (Easy, Medium, Hard, Very hard), which lets us see not only *how much* fine-tuning helps but *where* it helps.

a) *Static-camera caveat:* All experiments use the *static* (fixed-viewpoint) render, so the difficulty ablation is a like-for-like comparison under one camera. A side effect is that a small fraction of questions ($\approx 11\%$) refer to an object or event that the fixed viewpoint does not capture—it happens off-screen or behind an occluder—making them effectively unanswerable from the static view; this is a matter of camera coverage, not reasoning. Rendering the same scenes with the *moving* (orbiting) camera would recover many of these cases and is left to future work.

A. Overall Model Performance

Table VII reports overall accuracy for all three models across the five conditions and both prompting settings, and Figure 3 shows the complete picture broken down by test-question difficulty. Three findings stand out. First, **zero-shot physical reasoning is weak**: without fine-tuning, the models score between 17.8% and 32.9% overall—at or near chance for the multiple-choice format—confirming that off-the-shelf VLMs do not solve PHLOP. Second, **LoRA fine-tuning closes most of this gap**: the best configuration reaches 91.2% for InternVL3, 88.3% for SmolVLM, and 64.0% for the more brittle Qwen2-VL. Third, the **best curriculum is model-dependent**: SmolVLM peaks under the all-tasks (*full*) curriculum, whereas InternVL3 and Qwen2-VL peak marginally under *easy+medium* (92.0% and 68.2% with hints), with *full* a close second. The difficulty breakdown in Figure 3 also reveals a counter-intuitive pattern: after fine-tuning, the *Hard* and *Very hard* tiers are often answered at near-ceiling accuracy (InternVL3 exceeds 99% on *Hard* and reaches 100% on *Very hard*), while the *Easy* tier lags (e.g. 71.7% for InternVL3 without hints). The human-oriented *difficulty* labels (reasoning depth) therefore do not track empirical model difficulty: the hardest-labelled tiers are dominated by templates the models learn to answer reliably, whereas some “easy” perception and counting questions remain visually challenging.

B. Curriculum Dependence

Since every model is tested on the same test set and the curricula differ only in their *training* difficulties, Figures 4 and 5 show how the choice of training difficulties reshapes accuracy across the difficulty tiers. Training on *all* tasks (*full*) gives the broadest coverage: its polygon is outermost or tied on nearly every axis for every model, and it dominates especially on the *Hard* and *Very hard* tiers. Narrow curricula transfer unevenly. A model trained only on *hard* questions becomes excellent on the *Hard* tier (e.g. SmolVLM 97.8%, Qwen2-VL 98.0%) but collapses on *Medium* and *Very hard*, so its overall score is mediocre. Conversely, *easy+medium* is a strong all-rounder—for InternVL3 and Qwen2-VL it slightly edges *full* overall—because the bulk of the test set falls in the easier tiers. The models also differ in robustness. InternVL3 generalizes well even from a narrow slice (trained only on *hard*, it still reaches 84.4% overall without hints). Qwen2-VL is the most fragile: trained on *easy* only without hints it scores just 20.0% overall—below its own 22.6% baseline—so a too-narrow curriculum can actively hurt a weak backbone. Averaged over the three models (Figure 5), the ordering is clear and consistent: $full \gtrsim easy+medium > hard > easy \gg baseline$.

C. Improvement over the Zero-shot Baseline

Figure 6 expresses each fine-tuned score as its gain, in percentage points, over the *same model’s* zero-shot baseline. Gains are large and overwhelmingly positive: the all-tasks curriculum improves overall accuracy by +55.4 (SmolVLM), +46.3 (Qwen2-VL), and +68.5 (InternVL3) points with

physics hints, and individual difficulty cells improve by up to +98–100 points (e.g. InternVL3 on *Very hard*). The few non-positive cells are informative: Qwen2-VL trained on *easy* only *without* hints regresses on the easier tiers (e.g. −11.1 points on *Easy*), reinforcing that a narrow curriculum can degrade a fragile model rather than help it.

D. Effect of Physics Augmentation

Figure 7 connects each model’s no-physics score to its physics-augmented score for every curriculum and difficulty tier; a teal segment is a gain, a red segment a regression. The benefit of physics hints is strongly model-dependent. Qwen2-VL gains the most—up to roughly +22 to +25 points overall on the *easy* and *easy+medium* curricula—suggesting it leans on the textual parameters to compensate for weaker visual grounding. SmolVLM and InternVL3 show smaller, mostly positive shifts ($\sim +3$ to $+7$ points under the stronger curricula). Crucially, the hint is not uniformly helpful: at the zero-shot baseline it is neutral or slightly harmful (e.g. −4.9 points for Qwen2-VL, −1.3 for InternVL3), because the hint deliberately lists parameters for *all* objects—including distractors—so a model that cannot isolate the relevant object is misled rather than helped. This matches the prompt design of Section III-B3: the hint supplies context, not the answer.

E. Cross-Model Consistency

Finally, Figure 8 summarizes how much the three backbones agree, plotting the mean accuracy and ± 1 standard deviation across models for each curriculum and difficulty tier. Two patterns emerge. The zero-shot baseline is uniformly low and tightly clustered—all three models are similarly poor before fine-tuning—whereas after fine-tuning the mean rises sharply but the spread widens, because the backbones adapt at different rates (InternVL3 and SmolVLM far outpace Qwen2-VL). The all-tasks curriculum yields both the highest mean and, on most tiers, a tighter spread than the narrow curricula, making it the most dependable choice across architectures.

V. LIMITATIONS AND FUTURE WORK

PHLOP currently supports rigid-body dynamics with detailed rule-based annotations. Future extensions should explore deformable bodies, fluid interactions, and differentiable physics engines, significantly broadening the benchmark’s scope and realism (Tung et al. 2023). While PHLOP emphasizes object states and parameterized properties, broader benchmarks such as PhysicsRW (Wang et al. 2024), and PhysReason (Liu & et al. 2024b) point toward evaluating reasoning under novelty, open-world generalization, and multi-step causal chains. Integrating PHLOP with robotic manipulation tasks and real-world scene generalization could provide further validation of physical reasoning capabilities. Finally, incorporating open-ended language generation and learned classifiers for automatic semantic annotation could also enhance interpretability and generalization, laying the foundations for more robust, physically grounded AI systems.

TABLE VII

OVERALL ACCURACY (%) ON THE HELD-OUT TEST SET, AS REPORTED BY THE OFFICIAL PHLOP EVALUATOR, FOR THE ZERO-SHOT BASELINE AND THE FOUR LoRA CURRICULA, WITH AND WITHOUT PHYSICS HINTS. COLUMN NAMES DENOTE THE DIFFICULTIES EACH MODEL WAS *trained* ON; EVERY MODEL IS *tested* ON THE SAME FULL TEST SET (ALL DIFFICULTIES). A PER-DIFFICULTY BREAKDOWN IS GIVEN IN FIGURE 3. **BOLD** MARKS THE BEST CONDITION PER ROW.

Model	Physics hint	Baseline	Easy	Easy+Med	Hard	Full
SmolVLM	No	32.89	65.68	72.72	60.50	84.36
	Yes	32.93	70.36	79.99	59.08	88.33
Qwen2-VL	No	22.64	19.99	46.01	37.30	51.64
	Yes	17.79	44.83	68.17	50.66	64.04
InternVL3	No	24.00	65.56	86.00	84.38	84.23
	Yes	22.68	68.29	92.03	88.48	91.16

Our experimental study also carries several caveats that temper the conclusions above. **(i) Parameter-efficient adaptation:** we use LoRA rather than full fine-tuning to remain within a single-A100 budget, so the reported ceilings are lower bounds on what each backbone could achieve. **(ii) Single-run estimates:** each model-curriculum-prompt configuration is trained once; we therefore do not report variance across seeds, and small differences between conditions should be read as indicative rather than definitive. **(iii) Observational comparison:** the three backbones differ in scale, pretraining data, and frame sampling, so cross-model gaps reflect those factors as much as architecture, and we draw conclusions chiefly from within-model trends (curriculum and physics-hint effects). **(iv) Question imbalance:** the automatically generated templates are skewed toward stationary-duration and onset queries, and the difficulty tiers are unevenly populated, so a single overall score can be dominated by a few categories; the per-difficulty breakdowns (Figures 3 and 4) mitigate but do not eliminate this effect. Addressing these points—full fine-tuning, multi-seed runs, balanced sampling, and a larger model pool—is a natural direction for future work, alongside examining how physics-aware inductive biases, structured taxonomies, and simulation-grounded data bridge the gap between perceptual recognition and physically grounded understanding.

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Fig. 3. **Fine-tuning accuracy across task difficulty** (official evaluator, % correct; higher is better). Rows are curricula (baseline plus the four training sets); columns are the overall score and the four test-question difficulty tiers; the left block is physics-augmented, the right block no-physics. Fine-tuning on all tasks gives consistently high accuracy across tiers, and the hardest tiers are answered at near-ceiling once fine-tuned.

Yi, K., Wu, J., Gan, C., Torralba, A., Kohli, P. & Tenenbaum, J. B. (2020), Clevrer: Collision events for video representation and reasoning, in ‘International Conference on Learning Representations (ICLR)’.

Zhang, K., Chen, R., Zhao, D. & Fan, Y. (2024), Contphy: A benchmark for physically consistent attribute-based scene understanding, in ‘Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)’.

accuracies are lower for the models that rely most on the textual hints, most visibly Qwen2-VL.

APPENDIX

For completeness, Figures 9 and 10 repeat the curriculum-versus-difficulty radar charts of Section IV for the *no-physics* prompting setting. The trends mirror the physics-augmented case—training on all tasks gives the broadest coverage and the hardest tiers are answered at near-ceiling—but absolute

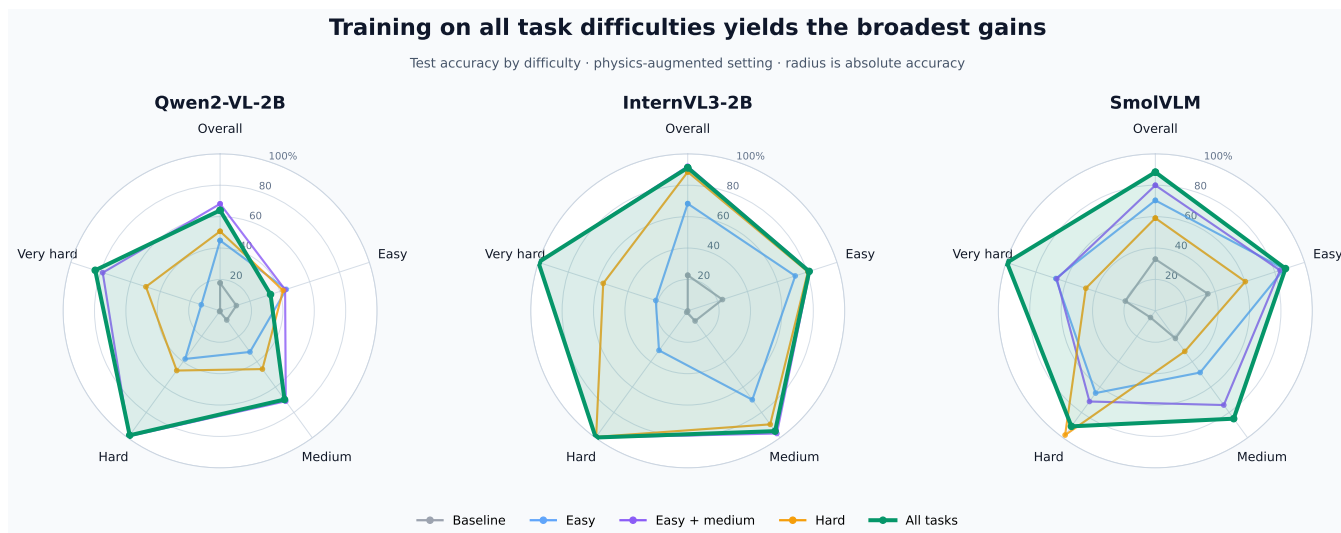


Fig. 4. **Per-model accuracy by difficulty (physics-augmented)**. Each panel is one model; the axes are the overall score and the four difficulty tiers; each polygon is a training curriculum. Training on all tasks (green) gives the broadest coverage. The no-physics version is in Figure 9.

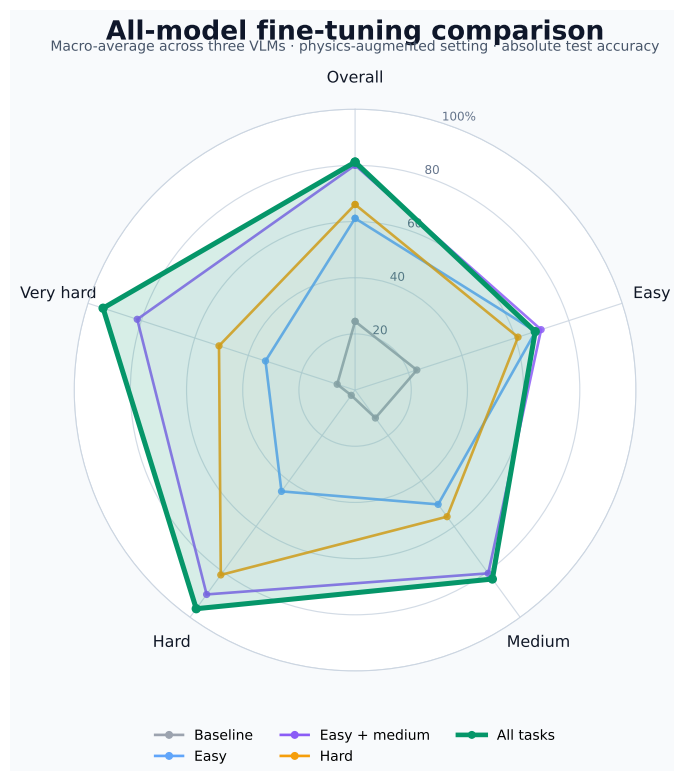


Fig. 5. **Macro-average across the three models (physics-augmented)**. Training on all tasks (green) is outermost on the harder tiers; *easy+medium* is a close all-rounder. The no-physics version is in Figure 10.



Fig. 6. **Improvement over each model's own baseline** (percentage points; green = improvement). Layout as in Figure 3. Almost every fine-tuned cell improves; the largest gains are on the InternVL3 backbone and on the harder difficulty tiers.

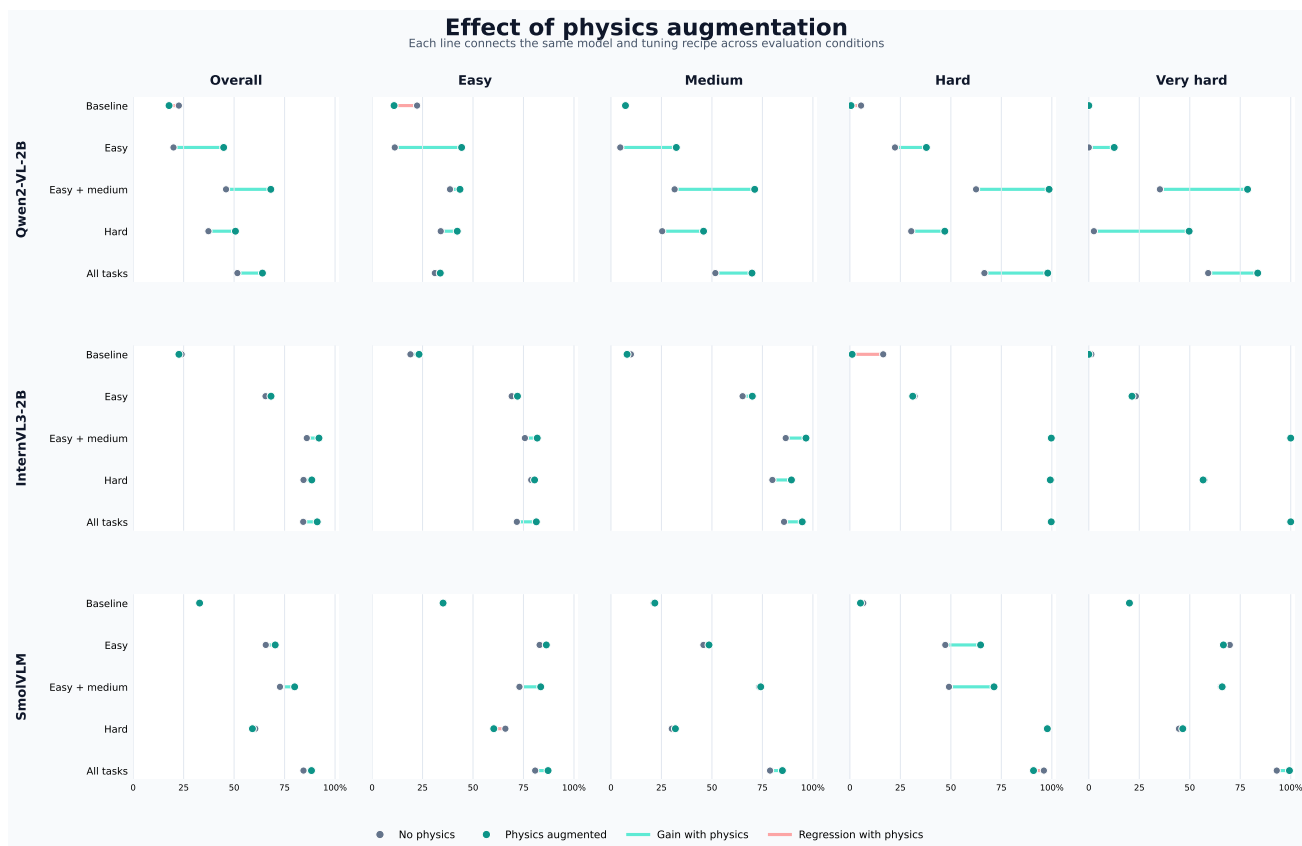


Fig. 7. **Effect of physics augmentation.** Each line connects the same model and curriculum without physics hints (grey) and with them (teal); teal indicates a gain, red a regression. Qwen2-VL benefits most; gains are smallest at the zero-shot baseline.



Fig. 8. **Consistency of fine-tuning across models.** Mean accuracy (dot) and ± 1 standard deviation (bar) across the three VLMs, per curriculum (rows) and difficulty tier (columns); the top block is physics-augmented, the bottom no-physics. Fine-tuning raises the mean but widens the spread; the all-tasks curriculum is both highest and among the most consistent.

Training on all task difficulties yields the broadest gains

Test accuracy by difficulty · no-physics setting · radius is absolute accuracy

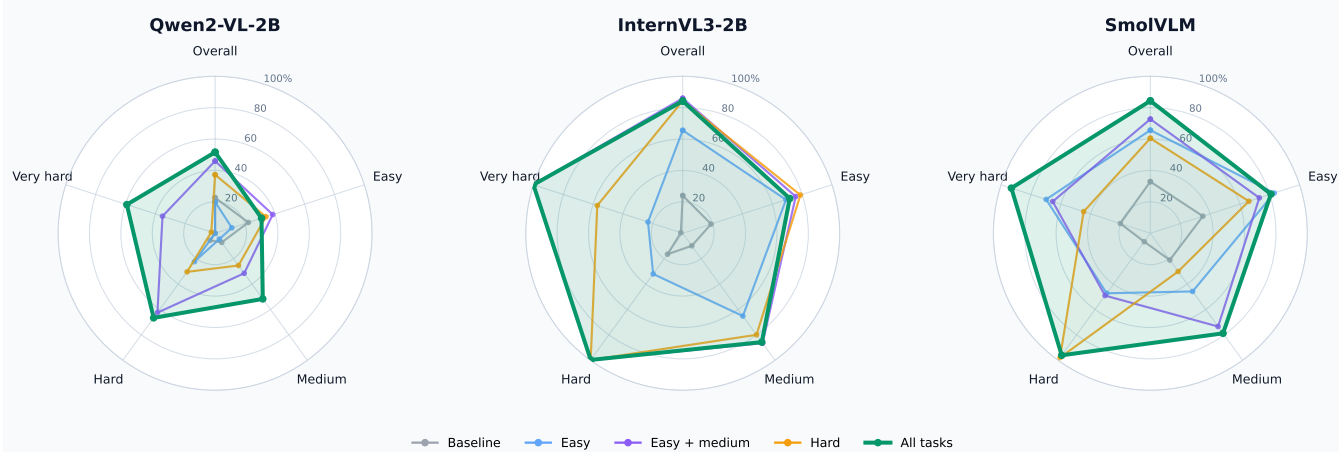


Fig. 9. Per-model accuracy by difficulty (no physics hints). Companion to Figure 4.



Fig. 10. **Macro-average across models (no physics hints).** Companion to Figure 5.